

ActInf GuestStream 073.1 ~ Lachlan Kent: "Mental gravity: Modelling the e

GuestStream | Lachlan Kent | 1:34:59

Hello and welcome. It is March 3rd, 2024, at least where I am. And we're here in active guest stream number 73.1 with Lachlan Kent, mental gravity, and I guess all that comes along with it. So thank you for joining, to you for a presentation, and then we'll have some questions from live chat. So thank you and to you. Thanks to you. My name is Lachlan Kent. I'm here in Melbourne. It's Monday morning at 6am. So I work at RMIT at Tech University here where I do very applied practical research into mental health promotion and prevention. And I think one of the major barriers to progress in my field is that we don't have a really solid scientific theory of what constitutes mental health and wellbeing.

And by contrast, how to define mental ill health for even the most common conditions like anxiety and depression. So I've been working for a number of years on a theory of mental wellbeing, effectively, based on the foundational and universal experience of gravity to try and get to the root explanation of why positive and negative mental states have a kind of vertical dimension to them. So depression, for example, is a painful downstate. Anxiety is a painful upstate. Well, joy is an upstate as well, but it's pleasant, not painful. And mindfulness or calmness are pleasant downstates. So I think there's kind of explanatory power in those distinctions when we look at how gravity shapes, biology, cognition and phenomenology. So I call it mental gravity, but I don't mean anything mysterious by it. It's just a way of modelling sort of physiological, mental and emotional states on their physical roots. All right, next slide. Thanks, Daniel.

Before I start, though, I'd like to acknowledge the unceded lands of the Boon Wurrung people of the Eastern Kulin Nation where I'm presenting from today. And I respectfully acknowledge their elders, past and present. And next slide again.

Okay, so today's talk, we're going to go through a few sections. We're going to start with a nice accessible anecdote, a parable. We're going to talk about Barbie, the Barbie film and Barbie's fall. And we're going to give a quick overview of the theory itself and then really focus on one of the key concepts and controversial concepts, I think, for mental gravity, the idea of a centre of mental gravity, and then relate that to active inference, sort of give a summary of where I come from and then take questions. Okay, so let's start with a kind of parable of mental gravity, Barbie's fall in the recent film starring Australian Margot Robbie. So maybe you've seen it, but even if you haven't, it's not a very complex premise for the film. So here we are at the start of the film, Barbie's sort of coasting along in her usual happy mode. So life's good. She has a strong sense of self. And the world is inherently predictable. She's happy. And this is depicted in the film as her floating kind of joyously sort of light and carefree.

But, next slide, thanks, Daniel.

But then the whole narrative arc of the film revolves around her fall. She starts uniformly positive. But then, there's no spoilers, this is in the trailer, she experiences an identity crisis.

So instead of being happy all the time, she starts feeling negative emotions. Something unpredictable happens that causes her to lose her sense of self. And on account of the loss, she experiences kind of an emotional breakdown. She says, this is the lowest I've ever been emotionally and physically.

If you could just click once there. And to really nail home the metaphor, when Barbie falls, these are the lyrics of the really beautiful melancholic song by Billie Eilish. It's,

I used to float. Now I just fall down. I used to know, but I'm not sure now what I was made for.

What was I made for? So it's an existential crisis for her that goes to the kind of core of her being.

She's not just sad because she lost something that can be replaced, or even someone that like dear to her, she kind of lost herself. And when people get depressed or have an identity crisis like this, we often say things like they're going through a mental breakdown. So depression affects hundreds of millions of people each year around the globe. It's kind of a universal phenomenon that needs scientific explanation, but we still really have no idea what we're talking about. Really. I think there's an intrinsic meaning here in the fallen status of depression that can fulfill a sort of much needed scientific explanation. You just click through to the next slide, Daniel, please.

Okay. Now here's a spoiler, although you won't be surprised, Barbie ends up kind of happy again at the end. She goes from really high. So it's sort of defying gravity to low where she's overwhelmed by gravity, gravity to somewhere kind of in between that's more balanced and more grounded where she's kind of managing gravity better. So both the song and the film end on a hopeful note. So Barbie recovers from her fall

when she rediscovers a new sense of self. And the song now goes, I think I forgot how to be happy. It's something I'm not, but something I can be. It's something I wait for something I'm made for. So mental gravity to me

is not just about falling. It's also about what people are for and what life is for. So we have to have ups and downs. They are a necessary part of life, but we all want the positives to sort of outweigh the negatives. That's another gravity based metaphor. We want to be up more than we're down. So mental gravity is also about how to recover from falls, or maybe how to prevent injuries from falling like depression. So it's really a theory of mental health and wellbeing of emotional positivity, of sort of human value of what we're made for. We just click through to the next slide, Daniel. Thank you.

Okay. So that's the pop psychology version. Now for the proper scientific explanation. The next slide. Thanks.

Right. So far in the last sort of 12 or six or 12 months, I've published two papers on mental gravity.

The first of which was pretty technical and focused solely on depression. The second of which is shown here, it's a bit less technical and more broadly focused on positive and negative mental health.

So it was in the journal of environmental psychology. And as you can see from the figure, or hopefully you can see in the figure, it framed the theory in terms of external environments at the top, internal environments at the bottom, and kind of intermediate socio-emotional environments in the middle. So we're going to take a quick tour through this summary figure. And let's kind of start not at the top, but in the middle, in the socio-emotional environment, where we found Barbie at a low sort of

mental and physical emotional state. You just click through a few times there, Daniel.

Okay. And then next slide.

Okay.

Okay. So the central idea in mental gravity is that there's a continuum of emotional states that range from up high and ecstatic to fallen down and depressed. So it's not a binary, of course, verticality isn't always positive. There are sort of intermediate states. So being up can also be a negative if you're unbalanced and say anxious about falling. And being down can feel good if you're balanced, stable and calm, like in mindfulness, meditation, or just regular relaxation. So these emotional states are a combination of physical states derived from complex interactions with the external environment and mental states derived from similarly complex reflective interactions within the internal environment. So there's a kind of mapping between physical and mental states that mental gravity seeks to explain. So assuming this is a human universal throughout cultures and evolution, how does this mapping come about? What are the neural structures that regulate it? And what does it mean to have gravity in or of the mind? Next slide. Thanks, Daniel.

Okay. So let's start with where it comes from. So the principle underpinning mental gravity is called cognitive gravitropism. Now gravitropism in plants and animals is the ability and kind of the drive to maintain alignment to and elevation within the gravitational environment. So for example, plants grow vertically,

animals have musculoskeletal structures to maintain verticality and balance, fish float using air sacks, birds fly with wings, etc. So these structures and functions that regulate verticality have positive adaptive value for survival and evolution. So maintaining gravitational uprightness and stability is a homeostatic function like temperature control or pH or nutrition, whatever. So if we fail to regulate our body's interaction with gravity, we die. It's as simple as that. Now cognitive gravitropism is where that positive adaptive and homeostatic value has been internalized and co-opted as a kind of mental template for value in other domains, namely socio-emotional expression or experience. It's kind of the value of the self in socio-emotional terms. So this template means that we think, talk, feel and act in ways that are also gravitropic because being mentally up and aligned is generally preferable to being down and misaligned. This effectively means that the mind has a vertical dimension that people strive to enhance through socio-emotional structures and functions. Thoughts, behaviors and emotional expressions kind of all act in service of a drive to optimize mental verticality. So this shows up in language as metaphor. In English, happiness is up, unhappiness is down, consciousness is up, unconsciousness is down, health is up, ill health is

down, more is up, less is down. So cognitive gravitropism is what makes these metaphors meaningful, and kind of universal because they also appear in other cultures and languages and even in pre-verbal infants. So mental gravity predates language in development at least. Next slide, thanks Daniel.

Which brings us to where gravitropism comes from. So we're at the top, we're in the external environment, so it's interactions with the external physical environment during childhood development. So gravity has been a constant through human evolution, it hasn't changed, but its effects are not constantly

felt throughout early development. So embryos begin life suspended in amniotic fluid, sort of floating in a state of neutral buoyancy. But this doesn't last long. As fetuses grow during pregnancy, gravity becomes more and more salient as their mass increases, their buoyancy becomes negative, I think, until finally the gravity support of their mother's womb is kind of taken away when they're born into a gravity laden world. So babies and children then have to learn to control their body under gravity. So mastering all kinds of postures and movements.

So children strive through effort to approximate that embryonic foundational effortless experience of buoyancy by maintaining increasingly complex states of verticality and balance as effortlessly as possible. So this learning involves a lot of falling, obviously, which is the opposite of buoyancy.

So there are complex neuroanatomical structures that help children adapt and adopt that sort of complex balance

function. So the vestibular system, which includes the posterior insular cortex, regulates verticality through gravoception to maintain bodily balance, control gait, and all the other physiological functions required to optimize bodily states with respect to gravity. And so up to this point, it seems like we're

just talking about physical states, right? Well, actually, no, not even at this foundational level, because gravoception, the sensing of gravity, the perception of gravity, also impacts higher order cognitive processes related to selfhood. So the feeling that we are our bodies, that we're embodied, is deeply rooted in the fact that our body has a center of gravity. So the egocentric perspective from which we view the world is anchored in the feeling that we can exert control over a distributed body weight.

But the mental doesn't actually stop there. So it goes to the next level. So next slide. Thanks, Daniel.

So by interacting with physical gravity, children also learn to simulate those interactions without feedback from the external environment. So you can just imagine yourself moving and the effects of gravity gravity on your body. Now, whereas physical gravity with feedback is mediated by the posterior insular cortex in the vestibular system, simulated physical gravity is mediated by the anterior insular cortex. And as you may know, the anterior insular is part of the salience network that regulates attention, interoception, affective responses to homeostatic signals, and most importantly for mental gravity, socioemotional expression. So this is how physical gravity states have been co-opted or mental gravity.

So the anterior insular cortex takes gravity that begins as an external physical and embodied physics and applies it then to the internal interoceptive affective environment. You can just click once there, Daniel, please.

So hopefully you can see a little pink ring around physical states. Is that where you're up to?

Yep. So that's where the physical states in the socioemotional environment derive from. That's how gravatropism has been copied from the external world and applied internally as a template for socioemotional meaning. And just click once more, thanks, Daniel. Or actually twice more.

But what about the mental states derived from the internal environment? Where do they come from?

So are they just interoceptive? Like the anterior insular? Well,

no, that's just the present state of the physical body. So Barbie didn't have a fever or have an upset stomach. She had an emotional reaction to an event that deviated from her long-term kind of entrenched

prior's expectations and world model. So the internal mental states must reflect long-term states of the self as derived from autobiographical memory. And this is where the rubber really hits the road, for mental gravity. Next slide. Thanks, Daniel. You'll click a couple of times.

So here we are, we're looking at the sort of top to bottom in the middle of the figure. So physical gravity states are the present state of the embodied self as regulated by the insular cortex. So both posterior and anterior. Mental states, however, are not restricted to the present and so are not physically embodied. So their source is from the past in the form of autobiographical or narrative sense of self. So it's not the what and where kind of embodied self, but the more diffuse who sense of self that is disembodied in a personal story or a narrative. So in the diagram, that's what's represented down the bottom in the internal environment. So that's life's ups and downs. And thanks to cognitive gravitropism, these mental states have adopted the structure of physical states. Just as it feels good to be physically up and stable in one's body, because it has a positive adaptive value, it also feels mentally good to have a higher and more stable sense of self. So self esteem, self worth, self regard. So these are the basis of mental health and wellbeing, because they are higher, more pleasant value states than their opposites. So health and happiness are up, ill health and unhappiness are down.

If you just click once there please, Daniel. So when Barbie's life story dips unexpectedly, she feels low in terms of self worth on a scale of verticality value, as I call it. And then click again, please.

And so she experiences a down state of having fallen, and once more. And she simultaneously feels low mentally and physically. Now to this point, none of this is counterintuitive or controversial, I don't think. Some of you might be thinking at this point, so what, this adds nothing explanatory because it doesn't give us sort of a causal mechanism. It just describes a process without really nailing down down how it happens. Well, this is where it gets possibly controversial, but definitely counterintuitive. This next bit is the stretch point. So next slide please, Daniel.

So my radical proposal is that the narrative or remembered sense of self also has a center of gravity, just like the physically embodied present sense of self. So think of it like this. We all agree that the mind has a structure. So memories accumulate, associate by learning and self organize into a complex network of nodes and connections. Well, if we take that structure literally and not just metaphorically, if we assume by a cognitive growth, gravatropism, that the structure has a vertical dimension. So some nodes are higher in the network than others, a hierarchy, then we can also assume that this structure can collapse. So if the network undergoes sort of rapid or radical reorganization, such as when Barbie encounters the shock of negative feelings, those high parts of the memory network structure may no longer be supported by the lower parts. So the center of gravity, the point at which gravity exerts vertical force on the network, will therefore fall. And the person will experience a lower mental state than they had enjoyed previously. Actually, Daniel, are we on the social emotional states? Is that what the heading is there? Cool.

So I realize that that may sound utterly ridiculous on first hearing. It's not something we're accustomed to thinking about. So we like abstract conceptualizations of the mind, not concrete ones like this. We like to imagine the mind as kind of a diffuse, ethereal,

and immaterial object that has no discernible physical properties, especially rudimentary physical properties like mass.

But I think the principle of cognitive gravitropism sort of forces us to change our conception of what constitutes the mind. So it's obviously a non-physical, immaterial, and purely mental object.

But that doesn't preclude the possibility that it has analogous physical properties.

In fact, if embodied cognition is true, then it makes sense to assume that the disembodied mental world retains some properties that mirror the physical world.

All right, next slide. Thanks, Daniel.

Okay, so let's look at the concept of the center of mental gravity in a bit more detail, because it all kind of hinges on this point.

Okay, next slide. Thanks.

So this isn't a new idea, and it's actually not mine.

So philosopher Dan Dennett in 1992, so he coined the term the self as a center of narrative gravity.

So he used this as a metaphor, but it was a serious metaphor to convey the idea that the self is not a mental object you can place kind of within a memory content,

but rather a point around which everything else revolves.

And by everything else, he meant autobiographical memories.

Okay, here's the analogy.

So the body's center of gravity is not a fixed location.

It depends on the body's posture and movement.

Well, the same goes for the sense of self.

It's not fixed in one place on one concept.

It's an abstraction.

It's a point in mental space.

And like the body's center of gravity, it can move depending on how the narrative is changing and what new memories are accumulating around that central point.

Okay, that's one way to conceive of the center of mental gravity in conceptual terms or philosophical terms.

Another is in neurobiological terms and the default mode network.

Just click once, please, Daniel.

So more recently, Christopher Davey and Ben Harrison from the University of Melbourne here in my hometown,

so talked about the default mode network as the brain's center of gravity.

So leaning on Dennett's idea of narrative gravity, they observed that the default mode network represents that narrative or even integrated sense of self.

So it generates a high order abstract representation of selfhood derived from autobiographical memory.

So it integrates memory to abstract out the central protagonist, the self.

And they also showed on a neurological level how the default mode network performs this integration by kind of placing itself at the center of the global connectome network.

And just click once there.

Thanks, Daniel.

So in another paper, these same authors put forward the idea that the self exists along a dimension from experiential to narrative poles.

So the experiential self is basically homeostatic states as regulated by the salience network.

And the narrative self is autobiographical states regulated by the default mode.

So this connection between those two is very important for mental gravity in associating those gravity simulations performed by the anterior insula to the narrative processes performed by the DMN.

And next slide.

Thanks, Daniel.

So this idea of centrality of the DMN is backed up by many neuroscientific studies of structural and functional connectivity.

So along with the cognitive control network or central executive network, as I call it, nodes of the DMN are the most globally connected regions of cortex.

So this places the DMN at the top of cortical organisation because the nodes are most dissociated from lower level primary sensory motor cortex.

And just a click through there, Daniel, thanks.

So in terms of topological organisation, this means that the DMN is positioned at an extreme of the cortical hierarchy

because it processes self-related information that is detached or unrelated to immediate sensory input.

It processes long-term trends in memory and narrative.

And one more click, please, Daniel.

So as argued by Carhartt, Harrison, Friston, this placement of the DMN at the top and centre of the cortical hierarchy

means it's functionally central for free energy minimisation and the phenomenology of selfhood, the ego.

And next slide, thanks, Daniel.

So if the mind has structure, including contents that are arranged vertically and prone to falling, then it makes sense to also speculate that memory has a kind of mental mass upon which gravity is exerted.

Just one click through, please, Daniel.

And so that's where the anterior insular cortex comes back into the picture.

So it is a key node of the experiential self because it sits atop the hierarchy of neural processes that extend from the peripheral nervous system

to the brainstem and subcortical regions.

They're making up the brain's central autonomic network.

But remember, it also regulates the simulation of gravity.

And so the assumption is that simulated gravoception acts upon an imaginary mental mass.

So Barbie kind of imagines her mental narrative self falling, even though her body's not falling.

But then, because she has imagined it, kind of self-fulfilling prophecy, maybe it's self-evidencing, I'm not sure, her body then follows suit and she feels physically low.

So gravity is part of how the present self interprets information derived from the narrative self in terms of

gravitational homeostasis.

So remember that embryos begin life floating in amniotic fluid.

So that is a foundational state of co-homeostasis inside the mother's womb.

That's referencing Anna Chianuka's work with Nick Wilkinson.

And like temperature and pH and all other homeostatic set points or attractors,

deviations from gravitational homeostasis yield negative affect.

So feeling heavy feels bad.

Feeling light feels good.

So Barbie begins the film floating through Barbie land, but then when she encounters negative affect for the first time,

she feels heavy and low physically and emotionally.

Not that she had a childhood at all, but her narrative has deviated from that embryonic foundational sense of self.

So her state of gravitational homeostasis is way outside her prior distribution,

but it is not physical gravity that caused her to lie down.

It was mental gravity.

And next slide.

Thanks, Daniel.

Okay.

So that brings us to the second last section.

So the relation to active inference.

So that idea of violated gravity priors brings us to the free energy principle and active inference.

I'm not an expert at all, but I think there are some connections that can be made between mental gravity and themes within the active inference field.

And next slide.

Thanks.

So the tenants of mental gravity map nicely onto the basic heuristic of active inference.

So just one click there, please, Daniel.

So external states would map onto the bodily interactions with the external gravitational environment, centred on the vestibular system and posterior insular cortex.

and just one click through. Sensory states would map onto the activity of the anterior insular cortex, the experiential and embodied self, which combines homeostasis and affect

with simulated gravoception. And next click. Thank you. So internal states would map onto the default mode activity, so the inner workings of the mental narrative self with its centre of

mental gravity. And one more click. And active states would be the combination of sensory and internal states by a socioemotional expression in thought, language and behaviour. So they're

the descriptive similarities, but I think there are deeper conceptual parallels as well, relating to that radical idea of a centre of mental gravity. And next slide. Thanks, Daniel.

Yes. Yep. Thank you. So this study by Kang et al. exemplifies what I mean in free energy terms about the centrality of the DMN. So they looked at energy landscapes of brain states using, estimating using pair, pairwise maximum entropy of resting state fMRI. So they identified multiple local minima, some of which mediate multi-step transitions towards the global minimum. So what is the global minimum? So it says here, coactivation of the prefrontal cortex and default mode network is captured as the global minimum. So as well as being central, the coactivation of the internally oriented DMN and externally oriented central executive network kind of represents the baseline of the brain's energy landscape. So in healthy or unpleasant states, healthy or pleasant states, the DMN and central executive network act together to maintain a sort of stable mental baseline. But what happens when that baseline becomes unstable or moves away from a homeostatic set point in pathological or painful states? Next slide. Thanks, Daniel.

So a recent study by Princeton and colleagues demonstrated that in depression, when I would argue the baseline

kind of drops, there is a shift away from the central executive network and towards the default mode and salience networks. So in conjunction with sensory networks and the cerebellum.

So the self has become even more central in an unbalanced way that people experience kind of as a fall away

from that external world and into themselves. So rather than feeling connected to the world by a sort of permeable boundary, it is as though their sense of self has become their entire world. So their past becomes all that there is, the future out there recedes away from a never ending past in here that they cycle through in

rumination. So next slide. Thanks, Daniel.

So that's what's depicted here in this figure in my earlier paper on mental gravity, with reference to the continuum of mental

gravity states. So up the top of the figure on the right, the self world boundary is kind of where it should be in positive

up or down states like joy and relaxation. So the salience network is paying at least some attention to the world.

And so there is a clear distinction between self and non self.

In depression at the bottom, however, when people feel down and low on account of a fallen sense of self, there's a sense in which the increased centrality of the DMN narrative self enlarges kind of the weight felt at their center of mental gravity,

energy, which then pulls the salience network, which is kind of at the boundary down and within that self world boundary.

So anxiety is included in that diagram in the middle, but I'm skipping over that at the moment.

So now gravity in physics is equivalent to space time curvature. So mass distorts space and time.

So I make the case in this paper that this literal depression of the mental self is akin to space time curvature, across all levels of phenomenological cognitive and neural dynamics.

And next slide. Thanks, Daniel.

Now, this is where for me, it gets very interesting for active inference and the free energy principle, because the physics of space time, the general theory of relativity, can be considered part of the gauge theory framework,

as argued by Sengupta et al in this paper from a few years back.

So mental gravity, if considered a form of curvature of neural, cognitive and phenomenological space time, could also be integrated within this gauge theory, neuronal gauge theory framework.

Just one click through, please, Daniel.

So consider this quote from their paper.

So crucially, the cognitive homologue of precision weighting in attention is attention, which suggests, suggests gauge fields are intimately related to exogenous attention.

In other words, attention is a force that manifests from the curvature of information geometry in exactly the same way that gravity is manifest when the space time continuum is curved by massive bodies.

So for me, there's a clear connection between mental gravity as a principle that connects external and internal environments via social emotional expression and mathematical frameworks underpinning active inference.

They're already kind of flirting with the analogy between space time curvature and physical gravity and psychological properties or processes.

And next slide. Thanks, Daniel.

So as an aside, before I finish up, I also think there is a way to implement space time curvature of general relativity in the brain in a way that is consistent with the tenets of mental gravity and space time curvature of cognitive and phenomenological fields, especially for depression.

So this is a work in progress with Dennis Lebihan.

We don't agree on everything, and he certainly doesn't agree that mental gravity is implicated here.

But he's the author of a relativistic model of the space time connect time.

So there's more to come on this front.

We'll keep working.

And that's kind of it.

That's a lightning tour of mental gravity.

So my theory of kind of what we are made for.

So the main objective is to provide a model of mental well-being, of being up and stable.

This is what I believe we are made for, according to gravitropism and gravitational homeostasis.

So you can call it what you like, human flourishing, happiness, meaning in life, however you want to phrase it.

So that's it.

Thanks, Daniel.

All right.

Thank you.

And thank you for you and the audience for bearing with some slight fun, as we discovered in minutes before beginning.

But well done.

Okay.

Well, people watching live can add any questions.

But while they are adding any questions in the chat, just how did you get to this project?

How did you get to Active Inference?

What just kind of led you to being here now?

So originally, the connection happened sort of in my undergrad.

I was studying psych and physics at the same time.

And we were doing sort of black hole physics in astronomy and astrophysics.

And we were talking about depressions in space time.

And then an hour later, we were talking about depressions in psychopathology.

And I think it was really a linguistic thing that got me first.

Oh, okay.

We're talking about very similar things here that people seem to feel down and dark and slow and heavy.

And we've just been modeling that mathematically in physics.

I wonder if there's a way to use that maths to model something about brain structure or function.

And then during my PhD, I connected with Jakob Hovi here in Melbourne, who's doing self-evidencing.

And I think studied under Carl back in the day.

And so he introduced me to Active Inference and predictive processing.

We published a paper on that as part of my PhD.

And so I started thinking about, okay, there's something about priors and human experience of gravity that is probably expressed somehow in mental states, in mental well-being.

And that's kind of what drew me to it.

All right.

Cool.

Well, I'll look forward to any questions in the chat.

But I just kind of wanted to reiterate what I thought was one of the core points.

Posture is basic to moving animals and to mammals.

And you pointed out that that gravitational homeostasis is like pH or nutrition or sleep.

It's a fundamental moving set point situation.

If you want to get over a small fence, you might want to briefly have the distribution of your physical center of gravity move up.

You might plan out different routes, just like you might plan for your implicitly or explicitly, like your glucose levels to fluctuate.

And so that kind of skillful competency definitely connects to the fields and leaven work in the competency and generalized state spaces.

So then that kind of navigation plus gravitational homeostasis becomes co-opted analogously to how the linguistic does, like evidenced by your most recent answer.

But the gravitational homeostasis becomes co-opted as part of navigational, which is a cognitive or narrative degrees of freedom in generalized state spaces.

And there's just so many ways to go.

And it was awesome how you laid it all out.

So again, thank you for presenting.

Thank you.

Where do we go from the center of gravity Bayesian mechanics?

What kinds of older data sets or theories do you think are like worth revisiting?

Or what kinds of approaches does it open up?

I've worked with Paul Badcock, who works on the hierarchically mechanistic mind as well.

He's in Melbourne here.

So he's worked with Carl and others to define sort of evolutionary and developmental nested time scales for active inference.

And I think that kind of nesting of present experience within developmental experience, within cultural and evolutionary experience, I think is a way to approach it.

So Georg Northoff, who was maybe going to join us here today, but hasn't been able to make it.

So he would talk about nested time scales in even a theory of consciousness, that that nestedness is a necessary part of experience.

And deviations from nestedness lead to pain.

So those homeostatic set points are nested.

So I think there's work to be done there in the experience of time is really where it started for me.

The experience of time dilation in depression to me is analogous to gravitational time dilation.

So masses curve space time so that time slows down.

There's very good empirical evidence that in depression, people experience a slowing of their conscious experience.

So not like their internal clock is running slowly.

It's not like they misjudge times all the time.

I think there are really fine technical ways to elicit changes or distortions in their time perception.

But their subjective flow of conscious experience definitely slows down.

So in mania, the opposite of depression, that experience of time speeds up.

In anxiety, another upstate, it's also sped up kind of chaotically.

So I think there are ways to, I think, pull out, okay, what's your experience of time?

And then how do we sort of recover that?

How do we re-nest that through whether it's postural or I think maybe better to have, you know, mindfulness meditation that uses simulated gravity?

So asking people to imagine themselves floating.

So I do a lot of yoga and part of my practice is imagining the body light.

So that's all postural.

And it's trying to imagine all the movements.

Now, a bit like Tai Chi, that there's no weight.

You're just moving freely.

And so that seems to me to improve mood, I think, because that center of gravity, the physiological center of gravity, has then fed back on the emotional state, on the narrative sense of self, because they're connected.

The anterior insular doing that simulation, it's connected to the default mode.

And there's some feedback there.

I think time and sort of perceived imagined body weight are two ways to sort of attack the problem.

All right.

Awesome.

I'll go to a question in live chat.

Upcycle Club writes, how does the concept of mental gravity influence the perception of time and space in astronauts versus normal folk?

Okay, that's pretty complex.

So because they go from a state of regular gravity to microgravity in space and then back again.

So studies have been done showing that there's sort of visceral changes, visceral and physiological changes in space that lead to.

So again, it's not a change in perception of time or space.

So the perceptual field remains pretty unchanged.

But when people come back to Earth and they're sort of re-acclimatizing to gravity, there is a tendency to experience anxiety and depression.

Now, that could be related to the fact that they've just left the space station and they're back on Earth and they're feeling a bit, you know, a bit let down by it.

But there's ongoing work happening that I think they've just done actually some psychophysical experiments with the perception of time.

And they're not noticing, I don't think, anything major other than sort of the sensory motor stuff.

So people are having to, like when babies are born, they have to recalibrate how they move for gravity.

There's an adjustment period where people have to re-acclimatize and sort of retrain the sensory motor network to let go of the gravity expectations.

I'm not actually hugely conversant in that, but it's somewhere that I need to learn more about because the work that's been done.

Yeah, it's, it's formative, but yeah, we're getting there.

Cool.

One thing that that mention of the microgravity and the callback to the developmental part made me think of, and you referenced Anna C. Unica's work then, is like, in the single cell water environment, there's a neutral buoyancy.

So that means that all, so all subsequent gravitational homeostasis can be attributed to self and movement.

Whereas nutritional homeostasis has a weaning off of, and never a complete weaning off of, but gravitational homeostasis really remains, but also from the very beginning, as something that can be inferred reliably over evolutionary timescales, that it is under that mammal's agency.

Yeah, exactly.

Yeah, that's a universal constant.

And we're now pushing the boundaries of it by getting ourselves into microgravity where no animal other

than, I suppose, fish, you know, get neutral buoyancy.

And I think there's an argument that Nick and Anna make in their paper on sort of buoyancy in utero, that what we're trying to do all the time is approximate neutral buoyancy with posture and movement.

That it's the feeling of load that it's the feeling of load that is a deviation from that ideal state.

That's the prediction error that we're trying to minimise is the feeling of load.

And there's another, there's a person in America who's a gastroenterologist who's working on gravity in the gut and linking that to mental health conditions like anxiety and depression through irritable bowel syndrome.

And his idea is that sort of a failure of the gravity support mechanisms in the gut leads to, you know, changes in the gut microbiome that we then interpret as, you know, a change in G-forces.

It's like the gut is a G-force accelerometer, a bit like your vestibular system senses changes in the head.

The gut also has, you know, proprioceptive visceral senses for acceleration.

And that's why we say when we're nervous or anxious, we've got butterflies in the stomach.

It's like the gut has become light.

And that's from top-down predictions, a bit like Barbie saying, I've had this fall and now I'm lying down.

Your expectation when you're nervous is that you're under threat and that there's some change in the gravity acceleration that you're feeling in your gut.

So he and I are in conversation, so talking about how the gut-brain connection could be, you know, another way of exploring mental gravity as, you know, not just brain and nervous system or central nervous system, but also your peripheral nervous system and visceral feedback.

And that works for predictive processing because interoception is very top-down dependent.

Wow.

Makes me think of, yes, butterflies in the stomach and also heart or stomach sank like a stone, sank like a rock, which is to say, like, the prior experience, no resistance under the constraints of normal free-falling things.

It went down at this acceleration.

And to ask about that linguistic angle, like, are there any metaphors that aren't consistent?

It seems like in day-to-day conversation, it would almost be taken as a strange metaphor if something were inserted, kind of like Kandinsky, who asserted that warmer colors were associated with light and with higher notes of music and like cooler, slower, darker and lower notes.

And it was like, well, if that, when violated surprises people, then that speaks to the nature of that associative prior.

But I mean, are there any linguistic examples or anything that?

Not that I've come across.

No, because I assume that it's pre-verbal, that children have, you know, natural associations between higher space and, you know, certain characteristics.

So more, we have a metaphor that more is up.

So we're saying, you know, greater value, higher value is more things.

And babies will attend to higher space when there are fewer things than when there are lower.

So that violates their expectation.

They expect more detail in higher space.

So it's, yeah, I think it's universal across, in English, across languages, and because it predates language.

So I'd be very surprised if there was a contradictory metaphor.

The one that there are behavioral expressions that are kind of contradictory.

So I think of when people, say, win a tournament, so they've exerted all their energy and they win, often they'll fall down, they'll collapse in sort of joy.

And so that's kind of a behavioral contradiction.

So I think taking it too simplistically is going to be a problem because, you know, counterexamples will be easily found.

But I think there are ways to frame that in terms of mental gravity.

It's not just, it's a very complex interaction with the gravitational environment.

So lying down feels good sometimes.

So what they're expressing or what they're experiencing emotionally at that point is kind of being overwhelmed by the volume of emotion that they're feeling.

So it's still an expression of something gravitational.

It's just kind of inverted.

So I don't want to be too simplistic about it and say up is always good, balance is always good.

Because in play, in risk, moderate risk taking, falling is good.

So we've got to be very nuanced about the language that we use.

Great.

All right.

Question from Dean Tickle.

The response to being embedded in a gravitational influence setting seems to be quite distinct from being embodied.

So what does a buoyant mindset look like?

Always good?

Question mark.

Yeah.

I think it always feels good.

But whether that's always, so homeostatically feels good.

Whether allostatically that is always good, I'm not sure.

So if allostasis is about the prediction and anticipation, I always think of sort of MDMA at this point.

So there's a trade-off.

So people will take MDMA to get high and to have a super ecstatic, super buoyant.

It's almost like gravity doesn't exist.

They can dance for hours.

But then two days later, they're going to hit the wall and have a huge come down.

So I think it feels good in the moment because that's the anterior insula, homeostatically saying, this is great, this is cool.

But then the DMN is maybe going to anticipate and say, this isn't going to be great for our narrative.

There's going to be a longer term cost to this experience as well.

So, yeah, it might feel good in the present, but that doesn't mean it's going to feel good in the long term.

I think we've got Anna joining us, have we?

Yes.

Yeah.

Did that answer the question, though?

Did that get to the heart of that person's question?

I hope so.

Yeah.

Cool.

Anna, are you here?

Hey.

Anna, how are you going?

I'm okay.

How are you?

Well, thanks.

It's very late here.

I don't know what time is there, but it's like, it's Sunday evening.

We're in a three-sector conversation, which is, you know, equidistant, but also some of the most fun to coordinate, as we know.

Yeah.

Yeah, yeah.

Yeah.

Do you want to, like, say anything or ask anything, Anna, or we can continue talking or reading other questions?

No, no, no.

I don't have questions for now.

So, good.

Okay, cool.

Yeah, another thought or area was, you talked about up and down, but under a closed planet structural model, up is out and down is in.

It is gravity.

So, that's the central executive network.

That's the main mediator there.

And so, we're connecting with the external world that way.

Our default mode is our internal world, but we also have to imagine what other people have inside their internal world.

So, there's, you know, what's someone else's narrative?

So, that has to then connect with other people's up and down.

So, I think being able to then, so children start very egocentric and very, you know, they're not mind readers at that point.

So, then to be able to not only disembodiment their own sense of gravity or their own centre of narrative gravity, to then apply that to other people involves another turn.

Another, okay, that other person is out there and they have an internal world as well.

How do I engage with that?

And that takes then behavioural coordination in the external world.

So, behavioural control, empathy and self-control or compassion and self-control go very much together.

So, in that sort of relative space of multiple minds, multiple centres, I'm just centred in my own, but I need to connect with others.

I need to be able to regulate my behaviour.

If I'm manic and self-interested and everyone's up is my up, then that's dysfunctional, I think, or socially and emotionally not going to connect with people well.

That's why we kind of term it, I don't like saying pathological, but that's why we term it as socially dysfunctional and unsustainable in a way.

It's not, they're not embedding themselves in that social environment in a way that acknowledges that other people have different orientations, that their up might be my down.

Just to kind of like echo a part of that, when there are polar opposites, it's a polarised antipodal situation.

One's up as the other one's down.

But even someone standing relatively next to you has a partially diverging perspective, except when there's the construction of the flat playing field.

Like an ice rink or something that's been then secondarily engineered to be even flatter than the local curvature, like a linear accelerator for particles is extra flat, which allows extra measurement accuracy coordination.

But then as it goes further and further out, it becomes like a larger and larger project to maintain all of that.

Yeah, Anna.

Yeah, so since I read your paper, I said next time we're going to see you, I'm going to nag you with a question about this, because I wonder how much of this narrative between up is light and good and down is bad.

So it's coming from this like inherited, you know, inherited ontology where, you know, the light and God is somewhere in the, you know, the sky and the devil and hell is in the earth, right?

So basically we want to escape the hell and the death because we are all going to die and, you know, earth to earth, ashes to ashes, you know, we want to just like go up.

And that's good in that sense, right?

Because we're well, nobody wants to die.

And I think, you know, I love that because, you know, Pico de la Minarangola and all this Renaissance people, when they were fighting against the church was like stating that the earth was in the middle of the universe.

And actually everything was revolving around the earth and they were refusing the idea that, you know,

actually is the sun, it's not the earth.

And then they were saying, well, if the hell is in the middle of the earth and everything revolves around the earth, so everything revolves around the hell.

So that's kind of like ring the bell, ding, ding, ding, ding, you know, the heads of those people, like higher up in this, like, I think from now, moment on, I kind of like realized that, okay, so maybe there is something about the sun, which is like, you know, the heat and, uh, but my question to you is like, how much of this do you think is kind of like related?

it because in in a sense like the food is coming from the down I mean the food is coming from the earth right so everything that actually allows us to survive is not so as so don't you think that somehow this likes we have like some side effect of some very old ontology that we inherited where you know that is in the sky and it's good uh and it's you know and death is here and uh it's bad right with the earth but that's questionable because I said it's like if we don't have the carrots and the carrot is not coming from the sky it's coming from underneath us right so how how do you reconcile that so I think like being absolutist about it is a problem and saying that up is always good no matter what down is always bad no matter what I think even in our behavior or even in like mature approach to life a balance of up and down is required you don't want to be happy all the time there's no adaptive value in that you have to go through trials and tribulations so a kind of child like I want to grow up and I just want to go to the stars and you know be super light and be sort of divine I think is kind of you know counter counterproductive in a way and I think like we have words of being grounded and of being mindful that are very down oriented so I think there is a value in things that are here that are stable that are connected that are in balance like a lot of environmentalism is about humanity in balance with nature so that's possibly even more ontologically prior than monotheism and the idea that you know everything up is good yeah I agree with you but you know it's just like when um when I'm very tired right like for instance right now just like lying down is actually good yeah I just want to stop and lie down I don't want to go up right so so it's it's uh as you said it's like it's a balance yeah and we need to be careful not to smother in a sneak in without realizing some like from the kind of like you know dualist ontology it's like we have God and the things good in the sky and bad things are here which is yeah yeah there's it does just like the coping mechanism is I don't think that's that holds because we actually need to lay down yeah exactly actually half of our life is like we are lying down because you know we need to sleep yeah so yeah I think there's a kind of a way out of that sort of paradox and like in monotheistic religions they're talking about the afterlife in a sense like that's like if you had a bird's eye view of your entire life it is if you'd never gotten up if you'd never been vertical you wouldn't necessarily be you know classified as having lived so it's a necessary condition to get up but it's not sufficient it's not doesn't suffice the whole you know all the contents of that life so you can say well if the religious people want to talk about that then that's an afterlife reflection it's not a lived embodied way to be that is more about balance and the complex like we were saying the complex nuanced you know if I fall down it can it's like you can cry from laughter and

tears you can fall down from depression and ecstatic joy so it's about understanding those those complexities that makes it real world and not mythical so yeah in the paper I can't remember which one I say this is what mental gravity is not it's not about the fall of man it's not about you know some mythical sense it's not about quantum gravity it's just about trying to define and model all our complex and socio-emotional states using gravity as that prior experience the template yeah I'll ask a question from the live chat um upcycle club writes in terms of embodied information processing the brain processes information at different granularities how does this impact our conscious experience of mental gravity

well I look at uh sort of spatiotemporal granularity there so spatiotemporal scale so I think a lot about time and the nestedness of temporal scales so at the lower level you've got fine grain temporal scales so

we've got to make this kind of

For mental gravity, so when you've got centrality, you've got nestedness of those timescales.

They're sort of able to harmonise with one another, even through neural oscillations.

Like you've got, that's Georg Northoff's work,
which is looking at spatiotemporal alignment of those timescales.

So that's when you're feeling good and you're feeling not overly centralised.

I think that's when the power of the DMN becomes too overweighted.

A healthy or, you know, a light state of being is when there is alignment between spatiotemporal scales.

So, yeah, it's the difference, it's the connection.

So if there's a dimension of the self from anterior insula, and I should do it this way, anterior insula to default mode, then the nestedness of those scales is very important.

Wow, very interesting.

On like a little bit less of a neurobiological note,
that makes me think about estimating time,
estimating like double click speed

all the way on through seconds and minutes and hours and days.

And, you know, there's decades that take weeks and weeks that take decades, all that kind of discussion.

And a day on earth is like this long in this other thought.

And then those kinds of compression and dilation are used

to point towards the direction of a distribution.

It's kind of like saying, yeah, from this GPS coordinate to there,
you go this many east and that many north.

And then similarly in cognitive or narrative spaces,
it's like, oh, I was looking for this kind of album,
but a little bit more of that.

And from like this period or with faster beats.

Yeah, exactly.

So that kind of like rate and gravity connection,
it's also so concordant.

Yeah, it's very clear.

And there's Paul Badcock's work.

So the hierarchically mechanistic mind who did work with Carl.

So he's also done work on the social adaption theory of depression.

So there's mapping there between his thoughts about the nesting
of sort of evolutionary timescales through to, you know,
ontogeny and then fast neural timescales.

And the disturbance of that in depression.

So the social and emotional disturbance of that alignment in depression.

Depression is a way of sort of calling out to other people that you need help.

So Barbie is saying, I need other people.

I'm not just happy in myself.

I'm calling out to others.

I need support to get myself vertical again.

And yeah, there's a nice synergy there with Paul's work as well.

Yeah, that's awesome.

I was thinking of a few other, like you mentioned a person collapsing at the end of a sporting event.

And how that's a combination of like muscular fatigue and the desire to have the homeostatic set point be
down.

Because they want to be resting closer to sleep.

All these different things or the symbolic meaning of it.

And all that.

And then I thought about the float tank experience.

Yeah, yeah, yeah.

And fun house rides.

Amusement park rides like spinning.

Yeah.

And the dipping and then weight lifting as kind of an integral through time.

Potentially reducing the perceived weight.

Perceived load.

Yeah.

Through strengthening.

I mean, so how would you just say at these early days for the gravity research program, like how do you take a phenomena and just sort of, as you've probably done many times, like try to work through it in this gravitational way?

I think multidisciplinary approaches is the only way to go.

So I've got the guy in America doing gut brain stuff.

So gut gravity.

There's also Mark Whitman who I published a paper with on consciousness, on time consciousness.

He's very much into flotation tanks and using that to alleviate anxiety and depression.

I also think that psychedelics, there's something about going on a trip.

There's the phenomenology of, you know, expansive space and time that happens within a psilocybin trip that we can talk about that as the therapeutic angle is something about.

Space time curvature or the recentering of self and altered state of self, altered state of consciousness.

So I think there are multiple ways to look at it.

So, you know, different interventions, different conditions and different sort of emotional states that we're trying to, trying to model without, you know, forcing people into doing weightlifting and saying, it's all about this one monolithic way of doing it.

Like Anna was saying, like, it's not just about sacred spaces that tend to be expansive as well.

So churches and larger spaces, even forests, natural blue and green spaces have that expanse to them.

That's sort of therapeutic in a way.

I think there are multiple, multiple ways of looking at it.

It's like, I suppose it's like predictive processing.

That's a way of looking at things that you can apply to different fields.

I think mental gravity is something like that.

It's very foundational.

It's kind of central and implicit in many fields of work.

So I'm hoping to work with Mark on flotation, with Brennan in America on the gut brain stuff, with Anna about, and Nick on, you know, co-homeliostasis and the development of gravoception in utero.

I think there are many, many ways to look at it.

Awesome.

Yeah, Anna.

Thank you so much, Lachlan.

This is really, really interesting.

I have a question related to the, to the fact that, so I understand that verticality is super important, but don't you think that there's also that connected, verticality is connected with the fact, with the locomotion, with the fact that people can, when they are vertical, they can basically just like, you know, go there in the world and find food, you know, self-reported and make sure that.

And that's, that's, it's not necessarily because it's not enough just to stand, right?

No.

Yeah.

Vertical is like, if you're still, still standing vertically, it's not going to get you very far.

The same way if you stay still like laying down, it's all the same, right?

So how do you see this like verticality in relation to being static and being active?

So I think the active component is like related to motivational control.

So explore and exploit behaviors.

So being able to explore, if you're vertical, you have greater degrees of freedom of movement than when you're lying down.

So it's easier to get horizontal if you're vertical, but you also need a low center of gravity, need to maintain a balanced center of gravity in order to do that.

So more agility, more ability to move side to side is about maintaining a low center of gravity when you're vertical.

So they're kind of working in combination.

So I spoke to Nick Wilkinson about this and he's like, okay, it's when you're born, you're starting at this.

I can't do anything.

I, I can't move.

I can't get myself up.

I'm dependent on my caregivers for all exploration of the environment.

I can only see, I can't interact yet.

Then over time, as we develop that motivation to explore comes along with our ability to maintain verticality and prevent fall.

So we start by falling a lot, but that's because we haven't learned that that low center of gravity is what we need to regulate.

And I think it's the same for mental gravity.

You need to be vertical, but maintain a low center of gravity when you're vertical.

That's most sustainable and better for like exploring yourself, exploring, you know, your own narrative, if you're high and unstable, you may, might be in a, you know, manic or anxious or, you know, overly upstate.

That's not adaptive either.

So there's, I wouldn't want to say it's a U curve that it's, you know, it's there's an optimum state in between because it's really complex, but it's a bit like that.

There's a combinatorial way of approaching gravity.

That means it's not just about verticality and it's not just about low verticality or misalignment about horizontalness.

It's about a combination dynamics of, of the two.

That's an amazing point.

And I think points to kind of a key engineering trade-off, which is the bigger the building, taller the building, more further out the building goes, the higher you can see.

And then the higher the center of gravity becomes, which then under even small shaking of the land or like tower of Pisa status that becomes unstable.

So then when skyscrapers are built, there's a heavy weight to move the center of gravity down and to dampen and all these kinds of things that allows like taller narrative construction.

So that's super interesting.

And then I thought about insects and not just flying insects because there the verticality is not associated with sleep.

Like an ant nest mate will be active at a given elevation and then stop and rest, but they're not going down.

Whereas mammals, even ones that are on quadruped locomotion, they still do go down when they're cowering or like curling up.

But in the insects, they, they can't go outside of the ones that like do and all this stuff like that.

But just in general, a ground walking insect doesn't go down.

So that really isolates out the affordances and kind of possibilities for movement.

Because I think, and I made a great point that humans, the verticality becomes also associated with like being able to move basically.

So being down, it's like you're caught on flat feet.

It's not even just being standing.

You have to be like in a little bit more of a poised posture to react to basically anything, even to catch something very light, like thrown at you.

So that's very interesting how the phylogenetic priors, which go far deeper than any of our at least symbolically describable or understood cultural priors, which also are very relevant and recent.

But these kind of bilateral or going on land or having aqueous components, those are even deeper.

And like you said, they're pretty verbal.

Yeah, for sure.

Yeah, yeah.

Please.

I think this version of mental gravity is uniquely human.

I think.

I don't know if we can generalize it.

I think there might be different homeostatic ways for those animals are talking about it.

But I think this, you know, our up-down associations, I think a pre-verbal sort of, you know, non, you know, they're pan-cultural, but I don't think they're interspecies necessarily.

I think it may be for fish.

There's something about neutral buoyancy as well, but for insects, for the birds, who knows?

But that might be one of the characteristics of their spatiotemporal consciousness.

Who knows?

Yeah.

Hmm.

Yeah.

Like for a fish, there's a possibly strong environmental correlation between like the density and the temperature and the light.

And so there's affordance in principle to move up, but then they find themselves in a niche that might be like seasonal, but still captures this kind of like density, light, nutrient envelope because that's their niche.

Yeah.

Yeah.

Oh yeah.

Ecology.

It really is like that.

And about finding the right level of, you know, fly when it's time to fly, but then not fly when it's not time to fly and all that.

Yeah.

So you can, you can do experiments with that in, you know, fish in a tank that you accelerate, you know, to different G forces and see how they behave differently and what associations they make with nutrition or not just buoyancy, but, you know, light, like you're saying, there are probably priors from, you know, evolutionary priors, phenotypic, phylogenetic priors in there that you can elucidate.

Yeah.

Wow.

That's very interesting.

And bacteria, other single cell in even just small tubes, there could be gradient centrifuge with, with constructed nutrient gradients.

That's awesome experiment.

On the active inference side, you showed the particular partition with the internal and external and the sense and the action states, the four states.

So the three states, the sense, internal and action are the ones that are defining the agent or the organism.

And so I was thinking like, what is a heavy sensory experience?

Well, heavy metal, a heavy cake, something like that.

And then what's a heavy action?

Possibly it doesn't need to be even any more literal than heavy deadlift, or it could be a very burdensome task.

That's hard to, to push and then heavy internal states.

So it almost has a heavy and light, not valence access, but just in this kind of multidimensional space of active inference agents and models, there is kind of a separate heavy and light for these three different parts of the particular thing.

So I think that also just speaks to the applicability.

It's not saying that those three states of the thing are like the same because they have to be welded together by mental gravity.

It's actually that gravity is like an analogy frame or a metaphor that at the very least can be used and is used normally.

And you could also invent speech things and just see which, which one sounds more normal, which one sounds more like a language model wrote it and just use never seen before gravitational language versus anti-gravitational language.

Just see what the effect is.

Maybe people prefer one or the other if it's like fantasy.

Well, there's work being done at my university at RMIT in Melbourne by someone who used prosody, so the upward and downward inflection of speech, so pitch modulation, to analyse speech of depressed and non-depressed people as a kind of biomarker.

So how people talk, the naturalness of their prosody is kind of, could be diagnostic in a way because it shows what sort

of emotional valence is being used when they talk up and down.

There's really a flatness of tone in depression that people can sense and people are expressing,

I think because of this, like you're saying, what's the metaphor?

It's kind of like cognitive load.

We think about depression as, you know, the self has become too salient.

There's too much self-referential information that's a drain on cognitive capacity because it in itself is cognitive load.

So their ability then to think about the external world is limited.

So I think there are, again, multiple ways to look at this and multiple ways to test it and intervene and, you

know,

for therapeutic benefit.

So, you know, I think prosody as a, what's the weight of behavioural expression?

I think there's posture, there's facial expression, there's prosody, there's heaps of ways of looking at active states as well.

So then to connect it to writing, first off, it's very fascinating that language models can, through the statistics of language, without using things like the prosody and the timing, just from transcripts learn so much about generating other realistic sounding transcripts.

So it's like, what information is in the prosody and the timing?

That's just very interesting, but it can be a very, I mean, it helps us think about what those kind of communicative techniques are.

Yeah.

And the symbolic basis of language is more of a latecomer than the kind of resonant nature of communication on these types of...

On that level, on that vertical dimension, I suppose you'd call it.

Yeah.

Yeah.

And so Nick Wilkinson, who's worked with Anna, talks, you know, he's very interested in working on sort of soft robotics.

So how do we train an agent not based on, you know, a desired state that we train forward, but based on like that co-homeostatic neutral buoyancy?

That's the set point, that's the attractor that you're deviating from.

And then all the implementations of that, regardless of physiology, is translatable then.

So that's what you're trying to approximate with standing, you know, quadrupedally is neutral buoyancy, but supported with different structures and functions.

So I think there's big application there, not just in terms of language models, but in terms of robotics as well.

Yeah.

All of the features of embodiment in environments that we know, like if it was a muddy environment or if it was a slippery one,

whatever the embodiment were saying you have to stay upright in this environment and move is kind of enough of a affordance to test and explore the features of that landscape.

If it has like gravel or whatever situation it has, that's kind of what it has to come back to.

Otherwise, like you said, to get up no more is to die.

Yeah, exactly.

Yeah.

Super interesting.

Anna, do you have any last comments or questions?

Yes.

So I do.

So first of all, I want to comment very quickly when you just said, because I think this is super important.

So there is right now we are in the middle of like some sort of like shift in the perspective.

And I think it's for the best in the sense, like instead of starting with something highly sophisticated from the other centric perspective, we'll look at basically how we're going to get there.

Yeah.

And then we see that basically, as I said, well, the body needs to be like, you know, embedded into environment in a certain state that have been designed by billions of years of evolution.

Right.

So it's not propositional attitude content that we can kind of like logically extract from some abstract formula.

Yeah.

Yeah.

It's something that is done through trial and error.

And actually, this is something that changes all the time because while we change the environment, we need to also change the posture.

And now it's like with all the pollution that we have right now in the world, maybe we're going to adapt and develop new organs or something.

Yeah.

So, so, so, so this, that's, I think that's, that's something very important, but I, I, I really liked the paper and I really liked the way you, you know, bring the, like the, um, the basis back in this, in the core, right?

Like, uh, in, in, in under the spotlight, but my question is to you.

Um, so, uh, that, that's, that's very beautiful way to, um, understand like psychopathy in terms of like, uh, I know heaviness.

And I was actually thinking that light and light means both light and sense, like not heavy, but light, like, and sense, like, you know, dark, not dark.

Right.

Uh, so that's, that's also beautiful.

beautiful but um so how you include basically the social component in it so the role of the other in terms of like this uh dynamic of the gravity because we are um interestingly enough people who are depressed or something they uh disengage also from social interaction so it's like some sort of like flatness uh and heaviness but um not just heaviness but also like disengagement from the active exploration of environment be it physical like you know going to the gym or social like going to meet friends yeah so what is the role of the other in your model for this like um up and down structure so i think that i'll go back to paul badcock's work about the social risk hypothesis of depression that it's a bit like okay you take mdma to have an extreme high knowing that two days later you're going to have an extreme low i think his argument if i have it correctly is that depression is kind of similar that it's it's actually a call out to other people to say i'm going to be horizontal i'm laid out i don't have any energy i need you to help me bootstrap and get back to being vertical but that can become pathological not by any sort of neural dynamics it's not a brain disorder in that sense it's a social disorder because that care is not forthcoming or it's not

um it's not appropriate or it's not effective so i mean we look at a lot of the social determinants of mental ill health that's the field that i work in really day to day is how do we promote and promote mental well-being and prevent mental health conditions by uh modern modifying risk and protective factors so that risk that i suppose the depressed person is taking is they're saying i'm the only hope that i have left i have none for myself is that someone else will come and bring me back into connection with the world i can't do it i've run out of energy it's like a the idea that depression is failed anxiety there's an anxious phase prior to depression of reaching for connections reaching for stability and then when that has failed then it's like i have nothing left so i think it's a way of de-pathologizing the individual de-stigmatizing individual and looking to their environment to say how is society how are other people letting this person down how have they been let down by the world around them so i think it naturally implicates others and take even though the individual is focused on the self the effort from you know a therapeutic angle needs to be from the external whether it's the natural you know uh green and blue spaces can be therapeutic to connect to the natural world or connect to the social world

yeah that's that's super important thank you and not necessarily sometimes we let people down but in most of the cases also we put people down you know it's like yeah put me down uh yeah and let me down yes the the the other can be for the better and for the worst right exactly yeah also on the gravity and the multi-player multi-agent strange attractors so there's like the three body problem binary stars all these asteroid belts all these different stable configurations or unstable configurations or chaotic or rhythmic all those different movements and then on the perspectival part there's the world at the center self at the center ego centric view which is our kind of undeniable object related visual system and then there's the kind of other centrism like the sun everyone on the earth will reference the sun as the center point

or there's kind of in principle like a omni-centrism but that's always alluded to as like kind of in principle true true but then it doesn't ever license any kind of perspectival analysis except hypocritically because a sensor has to be chosen even if it's like a purely homogeneous gas then the center of gravity is just the center of the space and so it's like there's so much there with um how diffuse the material is and then well the gas clouds that are really far away they have centers of gravity so just because the mist is very ephemeral at the time scale that we're moving doesn't mean that for like a redwood tree or for something like a bacterium it doesn't have a totally different materials property so we can't really say like how the niche is by itself it's not like the desert is just hot because it still might be too cold for some other adjacently living creature yeah exactly so there are meant there are gravity models of economics as well so it's not just like we're talking but we're extending the physics of gravity for the first time into mental the mental domain they've been using gravity models it's newtonian gravity it's how sort of cities or states trade with each other where you're analogizing mass to the size of the population and then the transfer of force is the transfer of goods or services or people or whatever so i think there's a way to apply this at different scales as well to say okay if society is coherent and nested and you know based on sort of topographical or topological features of that society you can start

to say well where are people going to fall down in this area here is where you're going to see an increase in anxiety and depression which is what we're seeing now we're seeing okay we're using metaphors like society is fracturing or you know there's this over centralization of power in you know the top you know the one percent of the population so the people at the bottom are struggling so i think there are ways to also take this the mental gravity idea and apply it at different social or societal scales as well awesome yeah well anna if you have any last comments or questions and then otherwise you can have the last word on like where we go or what do we do but anna first then lachlan no i don't i don't have any last words so i'm going to let back in the last thanks for joining anna thank you so i think um to sum up i think okay mental gravity this is one aspect of sort of bio gravitation that we can look at i think the conversations that i'm having with people sort of around the world in different uh domains in different fields there is a way to make you know kind of a discipline out of this or at least a network of people who are interested in the effects of gravity on whatever it is whether it's the gut whether it's the brain whether it's development whether it's psychopathology consciousness whatever so my call out is to people if you're interested and if you want to make contact i am thinking of trying to sort of become a node in that network and try and um bring people together people like nick anna is interested given his work he's going into robotics or he's interested in going into robotics so i think there are many different fields that can come together to say okay we're all talking about the same thing how can we sort of formalize it get some you know agreement and some principles and i'll be working from you know at least a point of a point of connection and discussion so that's hopefully where i'd like to see this go not just in theoretical you know advancement but in socially i think we can bring some of these ideas together and daniel thank you for having me thank you for contacting this has been great it's forced me to really get my ideas in order and be able to communicate them so thank you for the opportunity and to the active inference institute thank you thank you up up yep up up all right thank you see you all next time thank you bye thank you

Thank you.